

Database Case Study: E-Commerce High-Throughput Schema

G.V. Raman Global University • Computer Science & Engineering • Semester 3

Case Background: Flash Sale Inventory Contention

During peak seasonal sale events, standard relational transactions on inventory tables experience severe row-locking contention and high latency. This case analyzes how intentional de-normalization and Redis caching layers prevent checkout deadlock while preserving financial auditing integrity.